

Complete PC Setup Guide

Follow every step in order to get the app running on your computer.

Tech Stack used in this project:

Layer	Technology
Frontend	React 19, Vite, TypeScript, Tailwind CSS, shadcn/ui
Backend	Node.js 24, Express 5, TypeScript
Database	PostgreSQL + Drizzle ORM
Auth	Clerk (email + Google OAuth)
Charts	Recharts
PDF	jsPDF + html2canvas

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Step 1 — Install Required Software

Install the following three tools on your PC before anything else:

1a. Node.js (v20 or higher)

Go to <https://nodejs.org> and download the **LTS** version for your operating system (Windows / Mac / Linux). Run the installer and follow the prompts.

After install, verify it worked by opening a terminal and running:

```
node --version  
npm --version
```

1b. pnpm (Package Manager)

After Node.js is installed, open your terminal and run:

```
npm install -g pnpm
```

Verify:

```
pnpm --version
```

1c. PostgreSQL (Database)

Go to <https://www.postgresql.org/download> and download the installer for your OS. During installation:

- Remember the password you set for the **postgres** user — you will need it in Step 5.
- Leave the default port as **5432**.
- Also install **pgAdmin 4** (it is included in the installer — keep it checked).

■ *pgAdmin is a visual tool to manage your database. You will use it in Step 4 to create the database.*

Step 2 — Get the Source Code

You have two options:

Option A — Extract the ZIP file

1. Unzip the **fintrack-source-code.zip** file you downloaded.
2. Place the extracted folder somewhere easy to find, e.g. **C:\Projects\fintrack** (Windows) or **~/Projects/fintrack** (Mac/Linux).
3. Open your terminal and navigate into it:

```
cd C:\Projects\fintrack (Windows)
```

```
cd ~/Projects/fintrack (Mac/Linux)
```

Option B — Clone from GitHub

If you have already pushed to GitHub, run:

```
git clone https://github.com/YOUR-USERNAME/fintrack.git
```

```
cd fintrack
```

Step 3 — Install Project Dependencies

Inside the project folder, run the following command. This installs all packages for the frontend, backend, and shared libraries at once:

```
pnpm install
```

This may take 2–3 minutes the first time. You will see packages being downloaded.

■ *Make sure you are inside the project folder (the one that contains `pnpm-workspace.yaml`) before running this command.*

Step 4 — Create the Database

Option A — Using pgAdmin (recommended for beginners)

1. Open pgAdmin 4 from your Start menu / Applications.
2. Connect to your local PostgreSQL server (enter the password you set during install).

3. Right-click on **Databases** → **Create** → **Database**.
4. Enter the name: **fintrack**
5. Click **Save**.

Option B — Using the terminal

Open your terminal and run:

```
psql -U postgres -c "CREATE DATABASE fintrack;"
```

Enter your postgres password when prompted.

Step 5 — Create the .env File

In the root of the project folder, create a new file called **.env** (no other extension). You can copy it from **.env.example** which is already in the project.

Paste the following content into the **.env** file and fill in your values:

```
# PostgreSQL database connection
DATABASE_URL=postgresql://postgres:YOUR_PASSWORD@localhost:5432/fintrack

# Clerk Authentication keys (get from clerk.com — see Step 6)
CLERK_SECRET_KEY=sk_test_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
CLERK_PUBLISHABLE_KEY=pk_test_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
VITE_CLERK_PUBLISHABLE_KEY=pk_test_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

# Session secret — any long random string
SESSION_SECRET=my-super-secret-random-string-change-this
```

Replace **YOUR_PASSWORD** with the PostgreSQL password you set during installation.

■ *The .env file must be in the root folder of the project — the same folder that contains README.md and package.json.*

Step 6 — Set Up Clerk Authentication

Clerk handles user login/registration. It is free to use.

1. Go to <https://clerk.com> and create a free account.
2. Click **Create Application**.
3. Name it **FinTrack**.
4. Enable **Email** and **Google** as sign-in methods, then click **Create application**.
5. In the left sidebar, click **API Keys**.
6. Copy the **Publishable key** (starts with pk_test_) and paste it as both:
 - CLERK_PUBLISHABLE_KEY
 - VITE_CLERK_PUBLISHABLE_KEY
7. Copy the **Secret key** (starts with sk_test_) and paste it as CLERK_SECRET_KEY.

■ Keep your Secret key private. Never share it or commit it to GitHub.

Step 7 — Push the Database Schema

This command creates all the required tables in your database (users, categories, transactions, savings_goals, recurring_transactions):

```
pnpm --filter @workspace/db run push
```

You should see output ending with: **Changes applied**

■ If you see a connection error, double-check your DATABASE_URL in the .env file — especially the password and database name.

Step 8 — Run the Application

You need **two terminal windows** open at the same time.

Terminal 1 — Start the API Server (Backend)

```
pnpm --filter @workspace/api-server run dev
```

Wait until you see:

```
{"level":"info", "msg":"Server listening", "port":8080}
```

Terminal 2 — Start the Frontend

```
pnpm --filter @workspace/finance-app run dev
```

Wait until you see:

```
VITE ready in xxx ms  
→ Local: http://localhost:5173/
```

Step 9 — Open in Your Browser

Open your browser and go to:

```
http://localhost:5173
```

You should see the FinTrack landing page. Click **Create Account** to register a new account and start using the app.

All features available after login:

Page	What it does
Dashboard	Summary cards, charts, recent transactions
Transactions	Add/edit/delete income and expenses

Recurring	Set up automatic repeating entries
Categories	Manage income/expense categories
Savings Goals	Set and track monthly saving targets
Reports	Date-range summary + PDF export
Profile	Update name, currency, savings target

Step — Troubleshooting

Problem	Fix
pnpm: command not found	Run: <code>npm install -g pnpm</code>
Cannot connect to database	Check DATABASE_URL password, make sure PostgreSQL is running
Clerk error on login page	Double-check CLERK keys in .env — no extra spaces
Port 8080 already in use	Close other apps using port 8080 or restart your PC
Port 5173 already in use	Close other apps or kill the process on port 5173
pnpm install fails	Make sure Node.js v20+ is installed: <code>node --version</code>
push command gives error	Ensure the fintrack database was created in Step 4
Blank white page in browser	Make sure both terminals (API + frontend) are running